

Conditionals

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Two Types of Conditionals

Compare These Examples

1. If Shakespeare didn't write *Hamlet*, then someone else did.
2. If Shakespeare hadn't written *Hamlet*, then someone else would have.

Intuitively, 1 is true and 2 is false.

So it must be wrong to write both of them as $\neg A \rightarrow B$.

1. If Shakespeare didn't write *Hamlet*, then someone else did.
2. If Shakespeare hadn't written *Hamlet*, then someone else would have.

There are some common, if misleading, names for these kinds of conditionals.

- 1 is called an **indicative** conditional.
- 2 is called a **subjunctive** or **counterfactual** conditional

Ideally, we'd like a theory of how each works.

Material Conditionals

The material conditional is what the truth tables say is the conditional.

- We will write it as $A \supset B$
- This just means $\neg(A \wedge \neg B)$, i.e., $\neg A \vee B$.

One interesting hypothesis is that this is the right way to interpret English language conditionals.

Perhaps we should say that two interesting hypotheses are that the two kinds of English language conditionals are best represented as material conditionals.

- Hypothesis 1: If it were the case that A it would be the case that $B = A \supset B$.
- Hypothesis 2: If it is the case that A then it is the case that $B = A \supset B$.

The first of these is wildly implausible; the second is a bit more defensible.

Antecedent Strengthening

If indicative conditionals are material conditionals, then they should satisfy **Antecedent Strengthening**.

That is, whenever $A \rightarrow B$ is true, $(A \wedge C) \rightarrow B$ should be true.

But there are intuitive counterexamples to this.

1. If I win the lottery tonight, I'll be very happy.
2. So if I get hit by a car on the way home and win the lottery tonight, I'll be very happy

But the big problem with the material conditional analysis is what it says about false antecedents.

Note that if the material conditional analysis is right, then $(A \rightarrow B) \vee (B \rightarrow A)$ is a logical truth.

This does not always sound like a logical truth.

- If Brett went to Australia over break then he stayed in Ann Arbor, or if stayed in Ann Arbor, he went to Australia.

Strict Conditional

Here's a more plausible account of the conditional that avoids the last problem.

- *If A, B means Necessarily, A materially implies B.*
- In symbols, $A \rightarrow B = \Box(A \supset B)$

What does $\Box$ mean here? Probably something epistemic; like it is known that.

Antecedent strengthening is still valid on this picture.

- Assume every A world is a B world.
- Consider an $A \wedge C$ world.
- It's an A world, by the rule for true $\wedge$.
- So it's a B world.
- So $(A \wedge C) \rightarrow B$.

So if you were worried about that problem, it's still hanging around.

But A being false isn't enough for $A \rightarrow B$ any more.

So it fixes those problem.

This is better than the material conditional theory, but not perfect. And we haven't said anything about the subjunctive conditionals.

Antecedent Strengthening

Here is one of the implications we can use to test the strict implication theory.

1. $A \rightarrow B$
2. So, $(C \wedge A) \rightarrow B$

That is **valid** on the strict implication theory. Is it valid for ordinary, English conditionals?

Subjunctive Counterfactual Conditionals

1. If I had struck this match, it would have lit.
2. If I had dropped this match in water, and struck it, it would have lit.

If you think 1 is true, but 2 is false, then you think antecedent strengthening is **invalid** for subjunctive conditionals.

Future Directed Indicative Conditionals

1. If I strike this match tomorrow, it will light.
2. If I soak this match in water overnight, then strike it tomorrow, it will light.

If you think 1 is true, but 2 is false, then you think antecedent strengthening is **invalid** for future directed indicative conditionals.

It is a little trickier to get a counterexample here. There are some very unnatural examples though.

1. If Shakespeare didn't write Hamlet, someone else did.
2. If we've all been collectively hallucinating that there is such a play as Hamlet, when in fact there is no such play, and it certainly wasn't written by Shakespeare, then someone else wrote Hamlet.

If pushed I might say 1 is true and 2 false, but I'm really not sure.

Challenge

Three examples to see how well you're doing with these.

- Find your own counterexample to antecedent strengthening for subjunctive conditionals.
- Find your own counterexample to antecedent strengthening for future-directed indicative conditionals, i.e., indicative conditionals where the antecedent is about an event that may or may not happen in the future.
- Find your own counterexample to antecedent strengthening for past-directed indicative conditionals, i.e., indicative conditionals where the antecedent is about an event that may or may not have happened in the past.

Is the following speech ok.

If I'd dropped this match in water and then struck it, it would not have lit. But if I'd struck this match, it would have lit.

A lot of philosophers and linguists have argued that this sounds really weird. And this is going to be relevant, because on the best theory of why antecedent strengthening fails, it should be ok.

Counterfactuals

These play a central role in lots of fields, including

- Causation
- Explanation
- Evaluation
- Planning

The most important of these is causation.

There feel like there is a close connection between these claims.

- Causal event C caused effect event E.
- If C hadn't happened, E wouldn't have happened.

It's not a foolproof connection - maybe E would have happened some other way. But there is a link.

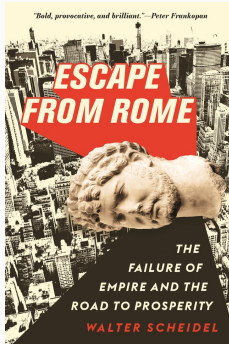


Figure 1: Walter Scheidel, *Escape from Rome: The Failure of Empire and the Road to Prosperity*

Part of what makes this provocative is that Prof Scheidel is doing what philosophers think is obvious, using counterfactuals to provide evidence about causation and explanation.

- But Scheidel isn't just interested in explanation.
- He's also interested in which facts about history are coincidences, and which are not.
- In particular, is it a coincidence that since the fall of Rome, there has not been a large empire in Europe that has lasted more than a few years.
- This is a striking contrast to the many empires that have arisen in South Asia, East Asia and the Americas.

Two Ways to Answer that Question

1. Is there something that could easily have happened such that if it had happened, there would have been a long lasting empire in Europe?
2. Is there a very similar world to ours in which there is a long lasting empire in Europe?

Intuition: These are sort of the same question.

1.1 Introduction

'If kangaroos had no tails, they would topple over' seems to me to mean something like this: in any possible state of affairs in which kangaroos have no tails, and which resembles our actual state of affairs as much as kangaroos having no tails permits it to, the kangaroos topple over. I shall give a general analysis of counterfactual conditionals along these lines.

Figure 2: Opening lines of *Counterfactuals*, by David Lewis, 1973

Minimal Change Semantics

- Replace the on/off accessibility relation between worlds with a distance measure d .
- So $d(x, y)$, where x, y are worlds, measures how similar x and y are.

If we're evaluating "If A were true, B would be true", we do the following:

- We find the nearest world, or worlds, where A is true.
- "If A were true, B would be true" is true just in case all the **nearest** worlds where A is true make B true.

If A were true, B would be true, means

- All of the nearest A-worlds are B-worlds.

From now on, I'll sometimes write this as $A \Box\rightarrow B$.

- This makes the conditional a variably strict conditional.
- It's strict, because it requires all A worlds to be B worlds.
- But it's variably strict, because which worlds it ranges over varies with what the antecedent is.

Here's the thought behind Lewis's view.

- When a counterfactual has a normal antecedent, the only worlds that matter to its truth are fairly normal worlds.
- To figure out what would have happened if I'd had bacon and eggs for breakfast, you only have to consider worlds that are really a lot like the actual world.
- But to figure out what would have happened if Columbus's fleet had sunk, you have to think about worlds that are really different to reality.

Complication 1: No A-world

We'll just stipulate that if there are no A -worlds at all, then $A \Box \rightarrow B$ is true for all B . These cases are weird, and I'll mostly set them aside.

Complication 2: Many equally closest A-worlds

Imagine that there are much of equally near worlds where A is true.

- And at some of them B is true, and at others B is false.

David Lewis argued that in that case both $A \Box \rightarrow B$ and $A \Box \rightarrow \neg B$ are false.

Could you have 1 and 2 false, but 3 true?

1. If a UM student had been elected mayor of Ann Arbor, it would have been an undergraduate.
2. If a UM student had been elected mayor of Ann Arbor, it would have been a postgraduate.
3. If a UM student had been elected mayor of Ann Arbor, it would have been either an undergraduate or a postgraduate.

Lewis said that you could have 1 and 2 both false, but 3 true, and that's why $\Box \rightarrow$ should work this way.

Another solution: Deny this is possible

But the other great founder of this tradition, Robert Stalnaker, argued that we should want these things to be equivalent (at least if $\Box \rightarrow$ was going to represent something in English).

1. $A \Box \rightarrow B \vee A \Box \rightarrow C$
2. $A \Box \rightarrow (B \vee C)$

And we couldn't have that on Lewis's picture.

- It is a constraint on the distance metric that there are no ties.
- The similarity measure is a strict ordering.
- And it's still a matter of debate whether Lewis or Stalnaker is right about this.

Logic of Conditionals

This fails on the minimal change semantics. Why does it fail? Because this is possible.

- The nearest A -world is 10 units away, and at it, B is true, but C is false.
- The nearest $A \wedge C$ -world is 100 units away, and at it, C is true (of course), but B is false.

Put in less technical terms, it fails when these things happen at once.

- In all normal worlds, when A is true, B is true and C is false.
- So there are no normal worlds where $A \wedge C$.
- But in the $A \wedge C$ -worlds that are only a bit abnormal, B is false.

Skiing Example

Here are some things we might imagine are true in all normal possibilities.

- When it snows, it snows a normal amount - it isn't a blizzard.
- When it snows, Jack goes skiing.
- Jack has no disposition to want to ski in a blizzard.

So in the most normal blizzard world, Jack does not go skiing. And we have both of the following.

- If it had snowed, Jack would have gone skiing.
- If there had been snow and a blizzard, Jack would not have gone skiing.

Transitivity (or Conditional Syllogism)

This argument seems initially compelling for any interpretation of 'if'.

1. If A , B .
2. If B , C .
3. So if A , C .

On the nearest possible world, or minimal change, semantics, this will fail for subjunctive conditionals.

- Imagine there is no nearby world where A .
- The nearest A worlds (which in these cases will be quite distant) are worlds where B is true and C is false.
- There are nearby, normal B worlds, and they are all worlds where C is true.

1. If there had been a hurricane and a blizzard on Presidents' Day Weekend, it would have snowed on Presidents' Day Weekend.
2. If it had snowed on Presidents' Day Weekend, Jack would have gone skiing on Presidents' Day Weekend.
3. So if there had been a hurricane and a blizzard on Presidents' Day Weekend, Jack would have gone skiing on Presidents' Day Weekend.

1. If I had crashed my car last week and won the lottery on the weekend, I'd have crashed my car last week.
2. If I had crashed my car last week, I'd now have less money than actually do.
3. So if I had crashed my car last week and won the lottery on the weekend, I'd now have less money than actually do.

Assume that you have *If A, C*, but not *If A and B, C*.

1. If *A* and *B*, *A*.
2. If *A*, *C*.
3. So, If *A* and *B*, *C*.

That should have true premises and a false conclusion.

That's the end of our little detour into conditionals.

We won't be putting these on the exam, but they come up so much throughout the rest of philosophy, and indeed everywhere else, that I thought it was worth doing a day on them.

Next time we'll start on probability.